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MEGaNorm: Integrated EEG/MEG workflows for multisite normative modeling of brain dynamics

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Summary

MEGaNorm is a Python package for large-scale analysis of electroencephalography (EEG) and magnetoencephalography (MEG) recordings. EEG records electrical potentials at the scalp, while MEG records magnetic fields generated by neural activity. Both help researchers study brain function. MEGaNorm combines preprocessing, automated quality checks and visual inspection, optional source localization, extraction of imaging-derived phenotypes (IDPs), integration of participant information, and normative modeling into a single workflow. The resulting models describe the range of brain measures expected in a reference population given age or other traits and quantify how each person compares to that range. This supports research on individual differences in brain function and precision neuropsychiatry. The workflow supports both sensor- and source-level analysis and runs on local computers or SLURM clusters. Source code, documentation, and archived releases are available through GitHub, Read the Docs, and the Zenodo concept DOI.

Statement of need

Normative modeling (Marquand et al., 2016) of brain measures depends on reference data that capture variation in the population of interest. To cover broad age ranges and diverse populations, EEG/MEG studies often pool recordings from several cohorts and sites. These datasets can differ in recording systems, sensor configurations, file organization, signal quality, preprocessing needs, and the format and coding of participant information. Researchers must also keep IDP definitions and participant identifiers consistent across these sources. Together, these requirements make multisite analyses difficult to set up, check, and reuse.

MEGaNorm addresses these practical needs for researchers using EEG/MEG to study brain development, aging, and differences between individuals or clinical groups. Its documented, configurable workflow reduces the need to rebuild scripts for cohort processing and normative modeling, while leaving researchers in control of processing steps, IDP definitions, and model design.

State of the field

Several pipelines already standardize or automate EEG/MEG processing. For EEG, PREP (Bigdely-Shamlo et al., 2015), HAPPE (Gabard-Durnam et al., 2018), and Automagic (Pedroni et al., 2019) automate preprocessing of large datasets, and DISCOVER-EEG (Gil Ávila et al., 2023) extends this to feature extraction from resting-state recordings. For MEG, FLUX

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(Ferrante et al., 2022) provides notebooks with explicit processing and analysis settings, including source analysis of oscillatory activity; MEGPrep (MEGPrep developers, 2026) and MEGAP (Mohammadi et al., 2025) automate preprocessing; and the ENIGMA MEG pipeline (Nugent et al., 2026) automates source-level spectral analysis of multisite resting-state data. MNE-BIDS-Pipeline (Jas et al., 2018) supports both modalities, covering processing from raw recordings to sensor- and source-level results, including evoked responses, time-frequency representations, and decoding.

For normative studies, a further task is to connect IDP extraction with cohort assembly, model fitting, and application to new recordings. Applying an existing model requires IDP definitions and processing settings compatible with those used for training. These connections span several libraries, so keeping them in a separate package lets the workflow be developed and reused as a whole. MEGaNorm provides this layer by linking recording-level processing, cohort data preparation, and normative analysis using PCNtoolkit (Rutherford et al., 2022). Its contribution lies in coordinating data, settings, and execution across these stages while drawing on established methods.

Software design

MEGaNorm separates IDP extraction and cohort assembly from normative modeling (Figure 1), allowing saved IDPs to be reused across modeling analyses without repeating signal processing. The `meganorm.src` modules implement signal processing, source localization, IDP extraction, and modeling; `meganorm.utils` handles configuration, files, participant data, and job management; and `meganorm.plots` provides plots for quality control, spectral inspection, and model evaluation. Processing settings and dataset layouts are kept separate so you can reuse analysis settings with different data sources. JSON configuration files define processing steps and parameters and are validated with Pydantic before processing begins. Dataset layouts specify paths and file patterns for recordings and participant information.

Each recording is processed independently using established analysis libraries. MNE-Python (Gramfort et al., 2013) supports preprocessing and source reconstruction. After preprocessing and quality control, IDPs can be extracted from sensor epochs or reconstructed source activity, grouped by brain region. The optional source step therefore precedes spectral extraction. SpecParam (Donoghue et al., 2020) models periodic peaks and the aperiodic background of power spectra. PyRASA (Schmidt et al., 2025) implements irregular-resampling auto-spectral analysis (IRASA) (Wen & Liu, 2016) to separate these components. Saved configurations, processing logs, and quality-control outputs document how researchers obtained the IDPs; they can also retain intermediate data, such as preprocessed recordings and power spectra.

Cohort assembly links the extracted IDPs with participant information in a single table for modeling. Researchers specify response measures, covariates, and batch-effect variables. MEGaNorm prepares these data for PCNtoolkit (Rutherford et al., 2022), which fits models, makes predictions, and evaluates performance. MEGaNorm provides plots of reference trajectories and individual deviations.

Local and cluster execution use the same core processing functions through different entry points. The `meganorm.API` layer provides a Python interface for sequential participant processing and local model fitting. SLURM entry points distribute recordings across independent jobs, allowing failed jobs to be resubmitted without rerunning successful ones. The existing modeling functions can also distribute fitting across response measures using PCNtoolkit's `Runner`. Separating processing configuration from scheduler settings lets the researcher reuse analysis settings across computing environments.

Tests and documentation support usage and maintenance of the workflow. Automated tests covering processing, IDP extraction, data handling, plotting, and model integration run through GitHub Actions. Sphinx documentation provides installation instructions, API descriptions, and worked examples.

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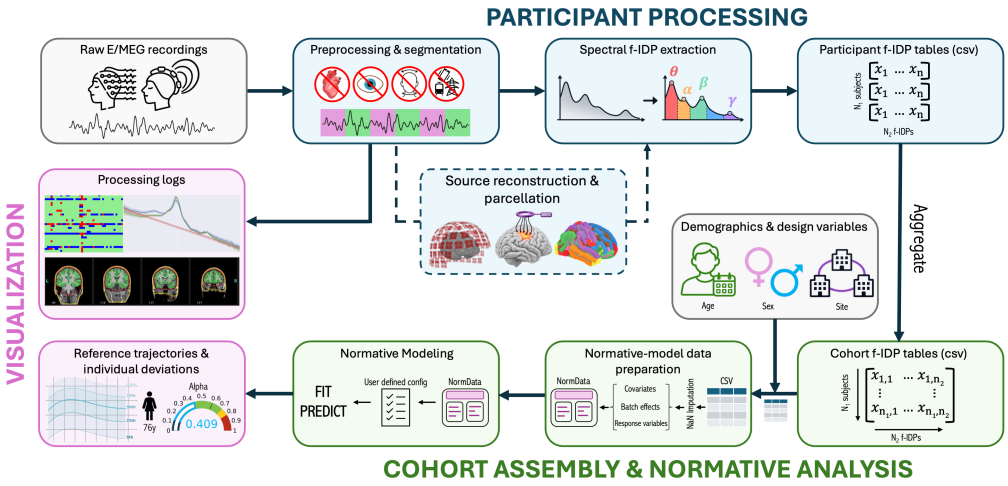


Figure 1: MEGaNorm workflow from EEG/MEG recordings to normative reference trajectories and individual deviations. Imaging-derived phenotypes (IDPs) are extracted from preprocessed sensor epochs or from reconstructed source activity grouped into brain regions (dashed path). Participant IDP tables are combined across the cohort and linked with participant information, such as age, sex, and recording site, for normative modeling with PCNtoolkit. Side panels show processing outputs and model visualizations.

Research impact statement

MEGaNorm supported the published study of normative MEG brain oscillations across the human lifespan (Zamanzadeh et al., 2026). The study used recordings from 1,846 reference participants aged 6-88 years, drawn from several datasets and recording systems, and applied the resulting models to a separate Parkinson's disease cohort. We used MEGaNorm for preprocessing, model training, statistical analysis, and figure generation. The Communications Biology article reports the scientific findings and validation for that study; the present paper describes the reusable software that supported the work.

AI usage disclosure

Claude Opus 5.5 and OpenAI GPT-6 assisted with software maintenance, test writing and troubleshooting, documentation, release metadata, and manuscript language editing. The human authors reviewed, edited, and validated AI-assisted work through code review, automated tests, and checks of technical claims and references. The authors made the scientific, methodological, and software-design decisions and take responsibility for the submitted work.

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References

Bigdely-Shamlo, N., Mullen, T., Kothe, C., Su, K.-M., & Robbins, K. A. (2015). The PREP pipeline: Standardized preprocessing for large-scale EEG analysis. *Frontiers in Neuroinformatics*, *9*, 16. https://doi.org/10.3389/fninf.2015.00016

Donoghue, T., Haller, M., Peterson, E. J., Varma, P., Sebastian, P., Gao, R., Noto, T., Lara, A. H., Wallis, J. D., Knight, R. T., Shestyuk, A., & Voytek, B. (2020). Parameterizing neural power spectra into periodic and aperiodic components. *Nature Neuroscience*, *23*(12), 1655–1665. https://doi.org/10.1038/s41593-020-00744-x

Ferrante, O., Liu, L., Minarik, T., Gorska, U., Ghafari, T., Luo, H., & Jensen, O. (2022). FLUX: A pipeline for MEG analysis. *NeuroImage*, *253*, 119047. https://doi.org/10.1016/j.neuroimage.2022.119047

Gabard-Durnam, L. J., Mendez Leal, A. S., Wilkinson, C. L., & Levin, A. R. (2018). The harvard automated processing pipeline for electroencephalography (HAPPE): Standardized processing software for developmental and high-artifact data. *Frontiers in Neuroscience*, *12*, 97. https://doi.org/10.3389/fnins.2018.00097

Gil Ávila, C., Bott, F. S., Tiemann, L., Hohn, V. D., May, E. S., Nickel, M. M., Zebhauser, P. T., Gross, J., & Ploner, M. (2023). DISCOVER-EEG: An open, fully automated EEG pipeline for biomarker discovery in clinical neuroscience. *Scientific Data*, *10*(1), 613. https://doi.org/10.1038/s41597-023-02525-0

Gramfort, A., Luessi, M., Larson, E., Engemann, D. A., Strohmeier, D., Brodbeck, C., Goj, R., Jas, M., Brooks, T., Parkkonen, L., & Hämäläinen, M. (2013). MEG and EEG data analysis with MNE-Python. *Frontiers in Neuroscience*, *7*, 267. https://doi.org/10.3389/fnins.2013.00267

Jas, M., Larson, E., Engemann, D. A., Leppäkangas, J., Taulu, S., Hämäläinen, M., & Gramfort, A. (2018). A reproducible MEG/EEG group study with the MNE software: Recommendations, quality assessments, and good practices. *Frontiers in Neuroscience*, *12*, 530. https://doi.org/10.3389/fnins.2018.00530

Marquand, A. F., Wolfers, T., Mennes, M., Buitelaar, J., & Beckmann, C. F. (2016). Beyond lumping and splitting: A review of computational approaches for stratifying psychiatric disorders. *Biological Psychiatry: Cognitive Neuroscience and Neuroimaging*, *1*(5), 433–447. https://doi.org/10.1016/j.bpsc.2016.04.002

MEGPrep developers. (2026). *MEGPrep: A reproducible Nextflow pipeline for large-scale MEG preprocessing* (Version 0.0.4). https://megprep.readthedocs.io/

Mohammadi, S. E., Shabani, H., Begmaz, M. M., & Dehaghani, N. S. (2025). MEGAP: A comprehensive pipeline for automatic preprocessing of large-scale magnetoencephalography data. *Psychophysiology*, *62*(7), e70109. https://doi.org/10.1111/psyp.70109

Nugent, A. C., Namyst, A. M., Carver, F. W., Thompson, P. M., & Stout, J. D. (2026). The ENIGMA MEG pipeline: Automated cortically localized spectral analysis of multi-site resting state MEG datasets. *bioRxiv*. https://doi.org/10.64898/2026.06.03.729876

Pedroni, A., Bahreini, A., & Langer, N. (2019). Automagic: Standardized preprocessing of big EEG data. *NeuroImage*, *200*, 460–473. https://doi.org/10.1016/j.neuroimage.2019.06.046

Rutherford, S., Kia, S. M., Wolfers, T., Fraza, C., Zabihi, M., Dinga, R., Berthet, P., Worker, A., Verdi, S., Ruhe, H. G., Beckmann, C. F., & Marquand, A. F. (2022). The normative modeling framework for computational psychiatry. *Nature Protocols*, *17*(7), 1711–1734. https://doi.org/10.1038/s41596-022-00696-5

Schmidt, F., Weisz, N., & Hartmann, T. (2025). PyRASA - spectral parametrization in python based on IRASA. *Journal of Open Source Software*, *10*(109), 7852. https://doi.org/10.

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21105/joss.07852

Wen, H., & Liu, Z. (2016). Separating fractal and oscillatory components in the power spectrum of neurophysiological signal. *Brain Topography*, *29*(1), 13–26. https://doi.org/10.1007/s10548-015-0448-0

Zamanzadeh, M., Verduyn, Y., Boer, A. de, Ros, T., Wolfers, T., Dinga, R., Šafář Postma, M., Marquand, A. F., Wingerden, M. van, & Kia, S. M. (2026). Normative modeling of MEG brain oscillations across the human lifespan. *Communications Biology*, *9*, 604. https://doi.org/10.1038/s42003-026-09825-2

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